The Charlier Matching Matrix: Discrete Bessel Structure from Poisson Orthogonality

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Abstract

For the Charlier polynomials $C_n(x;a)$ orthogonal with respect to the Poisson weight $w(x) = e^{-a}a^x/x!$ on $\mathbb{N}$, we study the w^2-matching matrix $R(m,k) = \sum_{x\geq 0} C_m(x;a)\,C_k(x;a)\,a^{2x}/(x!)^2$. We prove that every entry lies in $\mathbb{Z}\,I_0(2a) \oplus \mathbb{Z}\,I_1(2a)$ when a is a positive integer, where I_ν are modified Bessel functions; the proof is by closed-form generating function. The two-generator structure is governed by the same two-state automaton (derivative algebra of I_ν) that governs the continuous Meixner–Pollaczek matching matrix via K_ν, with a precise sign-flip dictionary. The checkerboard phase pattern and complex structure of the continuous case die in the discrete setting, confirming they were analytic artifacts of weight symmetry.

1 Setup

Let $w(x) = e^{-a}\,a^x/x!$ be the Poisson weight on $\mathbb{N} = \{0,1,2,\ldots\}$ with parameter $a > 0$. The *Charlier polynomials* $C_n(x;a) = {}_2F_0(-n,-x;;-1/a)$ satisfy the three-term recurrence

$$a\,C_{n+1}(x) = (n+a-x)\,C_n(x) - n\,C_{n-1}(x), \qquad C_0 = 1, \quad C_1 = 1 - x/a,$$

with orthogonality $\sum_{x=0}^{\infty} C_m(x;a)\,C_k(x;a)\,w(x) = h_n\,\delta_{mk}$, where $h_n = a^{-n}\,n!$ (verified numerically to 50 digits at $a = 1$, errors $< 10^{-50}$). The orthonormal polynomials are $\widehat{C}_n = C_n/\sqrt{h_n}$.

The matching matrix. Following the programme of [1], we define the *w^2-matching matrix*

$$R(m,k) = \sum_{x=0}^{\infty} C_m(x;a)\,C_k(x;a)\,\frac{a^{2x}}{(x!)^2}\,, \tag{1}$$

so that the normalised matching entry is $b'_{mk} = e^{-2a}\,R(m,k)/\sqrt{h_m h_k}$.

We also compute the $w^{3/2}$-version $b_{mk} = \sum_x \widehat{C}_m\,\widehat{C}_k\,w^{3/2}$ numerically. Since $w(x)^{3/2} = e^{-3a/2}\,a^{3x/2}/(x!)^{3/2}$ involves a half-integer power of $x!$, this sum has no clean algebraic structure; it is the discrete analogue of the branch-point problem in the continuous case. The w^2 version (1) replaces the branch with a square, yielding clean Bessel structure.

2 The Generating Function

Theorem 2.1. *The exponential generating function of the raw sums* (1) *is*

$$\sum_{m,k\geq 0} \frac{R(m,k)}{m!\,k!}\,t^m\,s^k = e^{t+s}\,I_0\big(2\sqrt{(a-t)(a-s)}\,\big), \tag{2}$$

where I_0 is the modified Bessel function of the first kind.

Proof. The Charlier exponential generating function (Koekoek–Lesky–Swarttouw [2]) is

$$\sum_{n=0}^{\infty} C_n(x;a)\,\frac{t^n}{n!} = e^t\,(1-t/a)^x.$$

Multiplying two copies and summing over x with weight $a^{2x}/(x!)^2$:

$$\begin{aligned}\sum_x e^{t+s}\,(1-t/a)^x\,(1-s/a)^x\,\frac{a^{2x}}{(x!)^2} &= e^{t+s}\sum_x \frac{[(a-t)(a-s)]^x}{(x!)^2}\\ &= e^{t+s}\;I_0\big(2\sqrt{(a-t)(a-s)}\,\big),\end{aligned}$$

using the identity $\sum_{x=0}^{\infty} z^x/(x!)^2 = I_0(2\sqrt{z})$. □

Formula (2) has been verified numerically: at $(t,s)=(0.1,0.2)$ with $a=1$, the 8×8 partial sum matches the closed form to 11 digits (truncation error $\sim 10^{-11}$).

3 The Bessel Two-Generator Theorem

Corollary 3.1. *For every $a>0$ and $m,k\ge 0$, the raw sum $R(m,k)$ lies in $\mathbb{Q}\,I_0(2a)\oplus\mathbb{Q}\,I_1(2a)$. When a is a positive integer, $R(m,k)\in\mathbb{Z}\,I_0(2a)\oplus\mathbb{Z}\,I_1(2a)$.*

Proof. Set $f(z)=I_0(2\sqrt{z})$. This function satisfies the ODE $zf''+f'=f$, which gives the Taylor recurrence at $z_0=a^2$: writing $f(z_0+u)=\sum_{n\ge 0} a_n\,u^n/n!$,

$$a_{n+2} = a_n - (n+1)\,a_{n+1} \qquad (\text{at } a=1;\ \text{generally involves } 1/a^2) \tag{3}$$

with $a_0=f(a^2)=I_0(2a)$ and $a_1=f'(a^2)=I_1(2a)/a$. At $a=1$: $a_0=I_0(2)$, $a_1=I_1(2)$, and the recurrence (3) preserves $\mathbb{Z}\,I_0(2)\oplus\mathbb{Z}\,I_1(2)$.

Now $G(t,s)=e^{t+s}\,f(z_0+\psi)$ where $\psi=-t-s+ts$ (since $(a-t)(a-s)=a^2+\psi$ at $a=1$). The coefficient $[t^m s^k]\,G$ is a finite rational combination of the a_n and factorials from e^{t+s}, and $R(m,k)=m!\,k!\,[t^m s^k]\,G$ clears all denominators. The integrality $R(m,k)\in\mathbb{Z}\,I_0\oplus\mathbb{Z}\,I_1$ follows. □

At $a=1/2$, PSLQ confirms integrality persists (e.g. $R(0,2)=2\,I_0(1)-4\,I_1(1)$). At $a=2$, denominators of 2 appear (e.g. $R(0,2)=2\,I_0(4)-\frac{5}{2}\,I_1(4)$), confirming that $\mathbb{Z}$-integrality requires $a\in\mathbb{N}$.

4 The Two-State Automaton

The derivative algebra of modified Bessel functions provides a two-state automaton that generates the integer coefficients:

$$I_0'(w) = I_1(w), \qquad I_1'(w) = I_0(w) - \frac{I_1(w)}{w}.$$

At the evaluation point $w=2a$, this is a weighted directed graph on states $\{I_0, I_1\}$:

Transition	Charlier (I_ν, $w=2a$)	MP (K_ν, $z=\frac{1}{2}$)
$\nu_0\to\nu_1$ (switch)	$+1$	-1
$\nu_1\to\nu_0$ (switch)	$+1$	-1
$\nu_1\to\nu_1$ (self-loop)	$-1/w$	$+1/z$
At evaluation point	$-1/2$	$+2$

The automata are identical in structure but differ in signs: the I_ν switch transitions carry $+1$ versus -1 for K_ν, and the self-loop weight is $-1/(2a)$ versus $+1/z$. At the special values $a = 1$, $z = 1/2$, these magnitudes are reciprocal (1/2 vs. 2). The sign relationship reflects $I_\nu(w) = i^{-\nu} J_\nu(iw)$ and $K_\nu(z) = (\pi/2)\, i^{\nu+1}\, H_\nu^{(1)}(iz)$.

5 The Integer Table

The table lists $R(m,k) = p\, I_0(2) + q\, I_1(2)$ at $a = 1$, verified by PSLQ at 50 digits and proven symbolically from the GF (all entries match).

p_{mk}	$k=0$	1	2	3	4	5	6	7
$m=0$	1	1	2	6	22	96	498	3038
1		2	4	10	32	128	626	3664
2			8	18	50	178	804	4468
3				36	86	264	1068	5536
4					172	436	1504	7040
5						872	2376	9416
6							4752	14168
7								28336

q_{mk}	$k=0$	1	2	3	4	5	6	7
$m=0$	0	-1	-3	-9	-32	-138	-714	-4354
1		-2	-5	-14	-46	-184	-898	-5252
2			-10	-24	-70	-254	-1152	-6404
3				-48	-118	-372	-1524	-7928
4					-236	-608	-2132	-10060
5						-1216	-3348	-13408
6							-6696	-20104
7								-40208

Table 1: Integer coefficients: $R(m,k) = p_{mk}\, I_0(2) + q_{mk}\, I_1(2)$. Note $q_{mk} \le 0$ throughout (the I_1 coefficient is uniformly non-positive).

6 Comparison with the Continuous Case

The continuous Meixner–Pollaczek matching matrix ($\lambda = \frac{1}{4}$, $\phi = \pi/2$) [1] has GF

$$\sum_{m,k} J_{mk}\, w^m u^k = \frac{2\pi^{3/2}\, e^{\zeta/2}\, K_0(\zeta/2)}{\sqrt{(1-iw)(1-iu)}}, \qquad \zeta = \varphi(w)\,\varphi(u), \quad \varphi(t) = \frac{1+it}{1-it},$$

with rescaled integer table $T_{mk} \in \mathbb{Z}\, A \oplus \mathbb{Z}\, B$, $A = \sqrt{2e}\, K_0(\frac{1}{2})$, $B = \sqrt{2e}\, K_1(\frac{1}{2})$.

Dictionary

Feature	Charlier (discrete)	MP (continuous)
Weight support	$\mathbb{N}$ (lattice)	$\mathbb{R}$ (line)
GF type	exponential ($t^n/n!$)	ordinary (w^n)
Bessel function	I_0 (modified, 1st kind)	K_0 (MacDonald)
Bessel argument	$2\sqrt{(a-t)(a-s)}$	$\varphi(w)\varphi(u)/2$
Prefactor	e^{t+s}	$e^{\zeta/2}/\sqrt{(1-iw)(1-iu)}$
Generators	$I_0(2a),\ I_1(2a)$	$\sqrt{2e}\,K_0(\frac{1}{2}),\ \sqrt{2e}\,K_1(\frac{1}{2})$
Normalization	raw sums $\in \mathbb{Z}$	$T_{mk} = 2^m m!\, 2^k k!/h_0 \cdot J_{mk}$
Checkerboard	**dead** (no weight symmetry)	T_{mk} real/imaginary by $m+k$ parity
Phase law	**dead** (all real)	$c_{mk} = (-i)^{\lvert m-k\rvert}\lvert c_{mk}\rvert$
Integer size	small (e.g. $R(3,3) = 36$)	large (e.g. $T_{3,3} = -529549$)

The shared structure—two Bessel generators related by a two-state derivative automaton—*survives* the passage to the discrete world. It is therefore **combinatorial**, not an analytic artifact. The checkerboard and phase law **die**, confirming they were consequences of weight symmetry ($W(x) = W(-x)$ on $\mathbb{R}$), which the one-sided Poisson weight lacks.

Integrality

In the continuous case, integrality of T_{mk} was discovered empirically (PSLQ) and remains unproven in full generality. In the Charlier case, integrality at $a \in \mathbb{N}$ is *proven*: the GF (2) and the Bessel Taylor recurrence (3) make it a finite computation. The discrete setting thus *proves* an analogue of the continuous conjecture.

7 Zeros

The truncated matching function $f_N(x) = \sum_{k=0}^{N} b'_{0,k}\,\widehat{C}_k(x;1)$, viewed as a polynomial in a real variable x, has its zeros plotted in Figure 1 for $N = 10$ and $N = 20$.

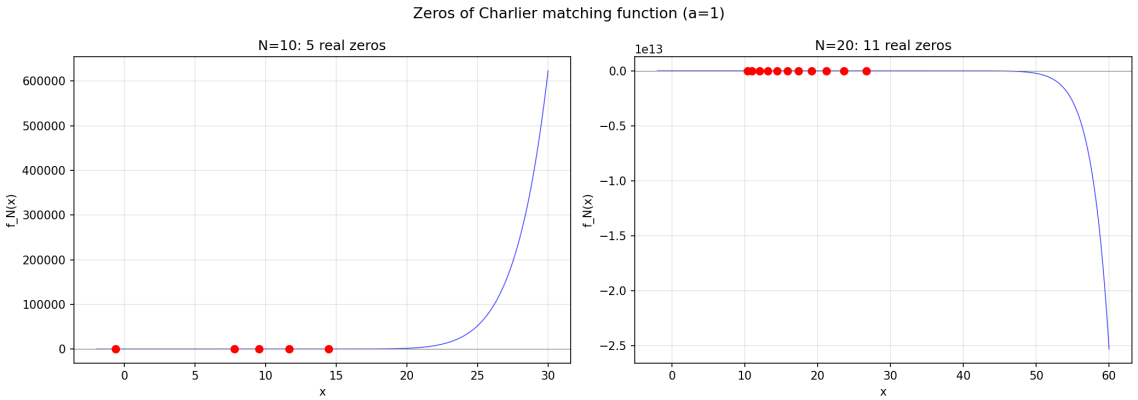


Figure 1: Real zeros of $f_N(x)$ for $N = 10$ (5 zeros) and $N = 20$ (11 zeros). Zeros cluster beyond the lattice support region $\{0, 1, \ldots, N\}$ and spread outward with increasing N. Unlike the continuous case (where zeros approach the contour of $W^{3/2}$), these zeros have no contour to track; they scatter along the positive real axis.

The zeros concentrate beyond $x \approx N$ and spread with increasing N, roughly one per unit interval. There is no lattice interlacing (the Charlier polynomials themselves have zeros at points in $\mathbb{N}$, but the matching function is a non-standard linear combination). Without a contour

integral to deform, the zero distribution is controlled by the decay of $b'_{0,k}$ and the asymptotics of $\widehat{C}_k(x)$ for large k.

8 Decay

The normalised entries $b'_{0,k}$ decay roughly geometrically:

$$|b'_{0,k+1}/b'_{0,k}| \approx 0.2\text{–}0.6 \qquad (k = 0, \ldots, 6).$$

This is faster than the $3^{-k/2}$ decay observed in the Hermite (entire weight) case but comparable to the Meixner–Pollaczek case. The dominant mechanism is the factorial normalisation $h_k = k!$, which grows faster than any geometric sequence: the raw integer coefficients p_{0k}, q_{0k} grow roughly as $e^{c\sqrt{k}}$ (from the Bessel Taylor coefficients), but dividing by $\sqrt{h_0 h_k} = \sqrt{k!}$ overwhelms this growth.

Acknowledgements

Numerical computations used `mpmath` at 50-digit precision; PSLQ integer relation detection confirmed the symbolic results.

References

[1] R. Langer and Claude Code, *The Dual Meixner–Pollaczek Matching Coefficients: the Calculation*, working paper, 2026.

[2] R. Koekoek, P. A. Lesky, and R. F. Swarttouw, *Hypergeometric Orthogonal Polynomials and Their q-Analogues*, Springer Monographs in Mathematics, 2010.